Objective:

To implement 4-bit parallel adder using VHDL

Architecture program

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity pa is
port (A,B: in STD_LOGIC_VECTOR(3 downto 0);
        S: out STD_LOGIC_VECTOR(3 downto 0);
        Cin: in STD_LOGIC;
        Cout: out STD_LOGIC
        );
end pa;
architecture behave of pa is
begin
process (A,B,Cin)
variable t: STD_LOGIC;
begin
t:=Cin;
for i in 0 to 3 loop
S(i)<= A(i) Xor B(i) Xor t;
t:= (A(i) and B(i)) or((t and A(i))or (t and B(i)));
end loop;
Cout<=t;
end process;
end behave;
```

Testbench:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity pa_tb is
end pa_tb;
architecture test of pa_tb is
component pa
Port ( A : in STD_LOGIC_VECTOR (3 downto 0);
 B : in STD_LOGIC_VECTOR (3 downto 0);
 S : out STD_LOGIC_VECTOR (3 downto 0);
 Cin : in STD_LOGIC;
 Cout : out STD_LOGIC);
end component;
--input signal
signal A,B:STD_LOGIC_VECTOR (3 downto 0):= (others => '0');
signal Cin:STD_LOGIC;
--output signal
signal S:STD_LOGIC_VECTOR (3 downto 0):= (others => '0');
signal Cout:STD_LOGIC;
begin
-- Instantiate the Unit Under Test (UUT)
UUT: pa port map(
A=>A,B=>B,S=>S,Cin=>Cin,Cout=>Cout);
process
 begin
 A<="0101";
```

```
 B<="0100";
 Cin<='0';
 wait for 20 ns;
 A<="1101";
 B<="0011";
 Cin<='1';
 wait for 20 ns;
 A<="1100";
 B<="1001";
 Cin<='1';
 wait for 20 ns;
 A<="1010";
 B<="1001";
 Cin<='1';
wait for 20 ns;
 wait;
end process;
end test;
```

Objective:

To implement 4-bit parallel Subtractor using VHDL

Architecture program

```
library ieee;
use ieee.std_logic_1164.all;
entity ps is
port(
	A, B: in std_logic_vector(3 downto 0);
	S: out std_logic_vector(3 downto 0);
	cin: in std_logic;
	cout: out std_logic
);
end ps;
architecture behave of ps is
begin
	process(A, B, cin)
		variable t: std_logic;
		begin
		t:=cin;
		for i in 0 to 3 loop
			S(i)<=A(i) xor (not(B(i))) xor t;
			t:=(A(i) and not(B(i))) or (t and A(i)) or (t and not(B(i)));
		end loop;
	cout<=t;
	end process;
end behave;
```

Testbench:

```
library ieee;
use ieee.std_logic_1164.all;
entity ps_tb is
end ps_tb;

architecture test of ps_tb is
component ps
port(
	A, B: in std_logic_vector(3 downto 0);
	S: out std_logic_vector(3 downto 0);
	cin: in std_logic;
	cout: out std_logic
);
end component;
signal A, B, S: std_logic_vector(3 downto 0):=(others=>'0');
signal cin: std_logic:='1';
signal cout: std_logic;
begin
parallel_adder: ps port map
(A=>A, B=>B, S=>S, cin=>cin, cout=>cout);
process begin
	A<="0101";
	B<="0100";
	wait for 20 ns;
	A<="1101";
```

```
        B<="0011";
        wait for 20 ns;
        A<="1100";
        B<="1001";
        wait for 20 ns;
        A<="1010";
        B<="1001";
       wait for 20 ns;
       assert false report "Reached end of test";
wait;
end process;
end test;
```